JARED KOLECKI

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EDUCATION

Ph.D in Physics **University of Notre Dame** **South Bend, Indiana**
Fall 2022 – Present
- Research focus in Astrophysics (Advisor: Lauren Weiss)

B.S. in Astronomy & Astrophysics **The Ohio State University** **Columbus, Ohio**
Fall 2017 – Fall 2020
- with Research Distinction (Advisors: Ji Wang, Jennifer Johnson)
- Minor in Computational and Informational Sciences

AWARDS AND HONORS

The Ohio State University
- Name and Seal Scholarship: 2020-2021
- Maximus Scholarship: 2017-2018, 2018-2019

CONFERENCES AND PRESENTATIONS

Talks

Stellar Oxygen Enhancement is a Predictor of Giant Planets July 2, 2026
Exoplanets 6 – Porto, Portugal: *Plenary Talk*

Planets are Made of Star Stuff May 1, 2026
Arizona State University: *TREC Coffee Hour*

Planets are Made of Star Stuff Mar 23, 2026
University of Wisconsin, Madison: *Monday Science Seminar*

Compositional Analysis of Planet-Hosting Stars Nov 17, 2025
Université de Montréal / McGill University: *Café IREx*

Graduate School: A Guide for First-Year Undergraduates Oct 18, 2022
The Ohio State Univsersity: Astronomy 2895

Posters

Building a Homogeneous Chemical Abundance Catalog of Planet-Hosting Stars Nov 6-7, 2025
Great Lakes Exoplanet Area Meeting – Madison, Wisconsin

Connecting Stellar Compositions and Planetary System Architectures Feb 3-7, 2025
Know Thy Star, Know Thy Planet 2 – Pasadena, California

Connecting Stellar Compositions and Planetary System Architectures Mar 16-21, 2024
Extreme Solar Systems V; Christchurch, New Zealand

REFEREED PUBLICATIONS (h-index = 7)

Full ADS Bibliography

First Author

Metal Pipe: A Broadly Applicable Stellar Abundance Pipeline Using Isochronal Parameters
Kolecki, J. R. and Weiss, L. M, *The Astrophysical Journal*, vol. 998, id. 341 (2026)

Measuring Elemental Abundances of JWST Target Stars for Exoplanet Characterization: FGK Stars
Kolecki, J. R. and Wang, J., *The Astronomical Journal*, vol. 164, id. 87 (2022)

Searching For Transiting Planets Around Halo Stars. I. Sample Selection and Validation
Kolecki, J. R.; Wang, J.; Johnson, J. A.; et al., *The Astronomical Journal*, vol. 162, id. 125 (2021)

Contributing Authorship

gr8stars II: judgement day for spectroscopic parameter model systematics
Freckelton, A. V.; Mortier, A.; Bedell, M.; et al., *Monthly Notices of the Royal Astronomical Society*, 549, 1070 (2026)

The Compositions of Rocky Planets in Close-in Orbits Tend to Be Earth-like
Brinkman, C. L.; Weiss, L. M.; Huber, D.; et al, *The Astronomical Journal*, vol. 170, id. 109 (2025)

A Reanalysis of the Composition of K2-106b: An Ultra-short-period Super-Mercury Candidate
Rodríguez-Martínez, R.; Gaudi, B. S.; Schulze, J. G.; et al., *The Astronomical Journal*, vol. 165, id. 97 (2023)

A Clear View of a Cloudy Brown Dwarf Companion from High-resolution Spectroscopy
Xuan, J. W.; Wang, J.; Ruffio, J.B.; et al., *The Astrophysical Journal*, vol. 937, id. 54 (2022)

Retrieving the C and O Abundances of HR 7672 AB: A Solar-type Primary Star with a Benchmark Brown Dwarf Wang, J.; Kolecki, J. R.**;** Ruffio, J.B.; et al., *The Astronomical Journal*, vol. 163, id. 189 (2022)

Elemental Abundances of nearby M Dwarfs Based on High-resolution Near-infrared Spectra Obtained by the Subaru/IRD Survey: Proof of Concept Ishikawa, H. T.; Aoki, W.; Hirano, T.; et al, *The Astronomical Journal*, vol. 163, id. 72 (2022)

TEACHING EXPERIENCE

Graduate Teaching Assistant **University of Notre Dame** **South Bend, Indiana**
Aug 2022 – May 2023
- Perform office hours, online help for undergraduate-level physics courses
- Grade exams, homework, and lab reports

Math and Science Tutor **Huntington Learning Center** **Lewis Center, Ohio**
Apr 2021 – May 2022
- Teach ACT and SAT math and science concepts and test-taking strategies
- Provide homework and conceptual help for students in high-school level math courses

Instructional Assistant **The Ohio State University** **Columbus, Ohio**
Aug 2020 – Dec 2020
- Assist students in weekly online labs, facilitating discussion and answering questions
- Grade student assignments

COMMUNITY WORK

PR Committee Chair **Graduate Physics & Astro Society (GPAS)** **University of Notre Dame**
Aug 2025 – Present
- Redesign and manage the GPAS website
- Use the site to promote student-led social/outreach events, maintain a list of useful resources

Social Committee Member **Graduate Physics & Astro Society (GPAS)** **University of Notre Dame**
Aug 2024 – Present
- Planning, set-up and tear-down for Physics & Astronomy graduate student social events

Physics Library Renovations **Physics & Astronomy Department** **University of Notre Dame**
May 2024 – August 2024
- Assist office admin staff with acquisition of surplus furniture/amenities, removal of old items
- Design and implement a floor plan suitable for academic and social uses
- Continue renovations where possible to create a more inviting space